Experiment 11

Title: Orchestration using Docker Compose & Docker Swarm (Continuation of Experiment 6)

PART A – CONCEPT CONTINUATION (Simple Explanation)

From Experiment 6, you already know:

Tool	What it does	Limitation
`docker run`	Runs a single container	Manual, no coordination
Docker Compose	Runs multiple containers together	Single machine, no auto-healing

New Concept: Orchestration

Orchestration = Automatic management of containers

Think of it like a restaurant manager:

- Decides how many waiters are needed (scaling)
- Replaces a sick waiter immediately (self-healing)
- Distributes customers evenly (load balancing)

What Orchestration Adds:

Feature	What it means
Scaling	Increase/decrease number of containers
Self-healing	Restart failed containers automatically
Feature	What it means
Load balancing	Distribute traffic across containers
Multi-host	Run containers across multiple machines

The Progression Path

```
  docker run → Docker Compose → Docker Swarm → Kubernetes
 |                |                |               |
Single container Multi-container    Orchestration   Advanced
                 (single host)      (basic)         orchestration
```

> This experiment focuses on: Moving from Compose → Swarm

PART B – PRACTICAL (EXTENSION OF EXPERIMENT 6)

Prerequisites

- Docker installed (with Swarm mode enabled)
- The docker-compose.yml file from Experiment 6 (WordPress + MySQL)

> Note: If you don’t have the file, here it is again:

```
# docker-compose.yml (from Experiment 6)
version: '3.9'

services:
db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
     MYSQL_ROOT_PASSWORD: rootpass
     MYSQL_DATABASE: wordpress
     MYSQL_USER: wpuser
     MYSQL_PASSWORD: wppass
 volumes:

       -  db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
       -  db ports: - "8080:80" restart: always environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
```

```
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
db_data:
wp_data:
```

Task 1: Check Current State (No Swarm)

First, make sure no containers from Experiment 6 are running:

```
# Stop any existing compose setup
docker compose down -v

# Verify no containers are running
docker ps
```

Expected: Empty list (or only unrelated containers)

Task 2: Initialize Docker Swarm

Swarm mode turns your current machine into a manager node of a cluster.

```
docker swarm init
```

What this command does:

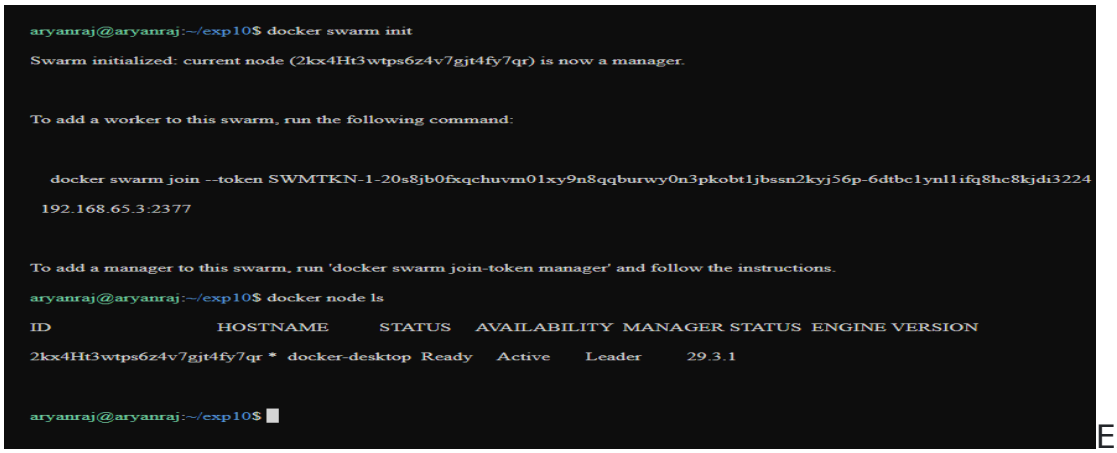
- Enables Swarm mode on your Docker daemon
- Makes this node a “manager” (can control the cluster)
- Creates a join token for worker nodes (not needed for single-node) Expected output:

```
Swarm initialized: current node (xxxxx) is now a manager.
```

Verify Swarm is active: `docker node ls`
Expected output:

```
ID                            HOSTNAME   STATUS    AVAILABILITY  MANAGER
STATUS
xxxxxxxxxxxx                   your-pc    Ready     Active
```

Leader


Explanation:

This shows your node is ready as a Swarm manager.

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Task 3: Deploy as a Stack (Not Just Compose)

In Swarm, we deploy a stack (a group of services) using the same Compose

file. `docker stack deploy -c docker-compose.yml wpstack`

Breaking down the command:

- docker stack deploy → Deploy a stack to Swarm
- -c docker-compose.yml → Use this Compose file
- wpstack → Name of the stack

What happens behind the scenes:

1. Swarm reads the Compose file
2. Creates services (not individual containers)
3. Services manage containers automaticallyExpected output:

```
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
```

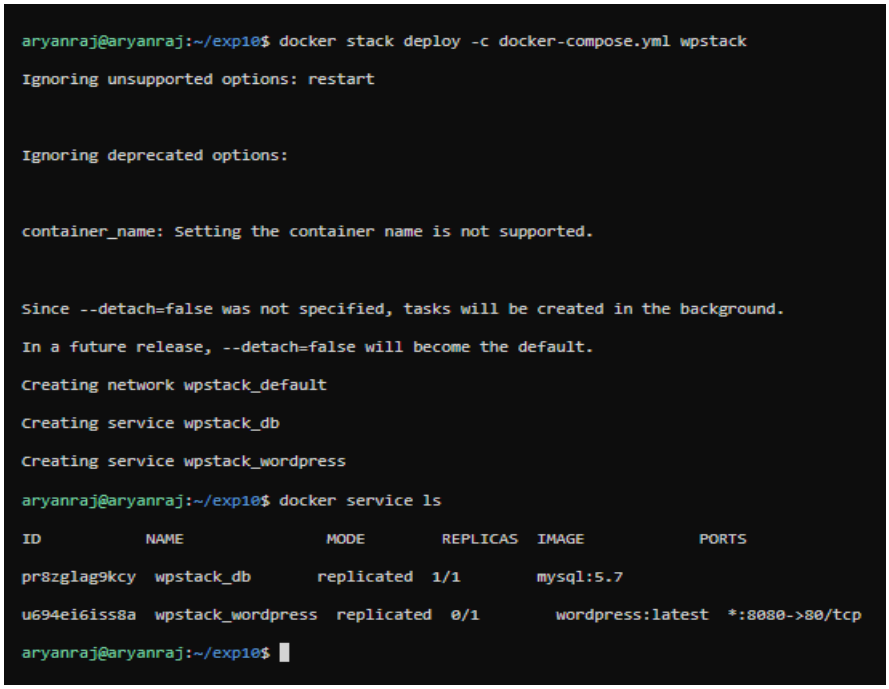
Task 4: Verify the Deployment

List all services in the stack:

```
docker service ls
```

Expected output:

```
ID             NAME                MODE         REPLICAS   IMAGE
xxxxx          wpstack_db          replicated   1/1        mysql:5.7
xxxxx          wpstack_wordpress   replicated   1/1        wordpress:latest
```



> Key change: Notice the names are `wpstack_db` and `wpstack_wordpress` (stack name + service name)

See detailed tasks (containers) for a service: `docker service ps wpstack_wordpress`

Expected output:

See all running containers:

```
docker ps
```

You'll see containers with names like:

```
wpstack_wordpress.1.xxxxx
wpstack_db.1.xxxxx
```

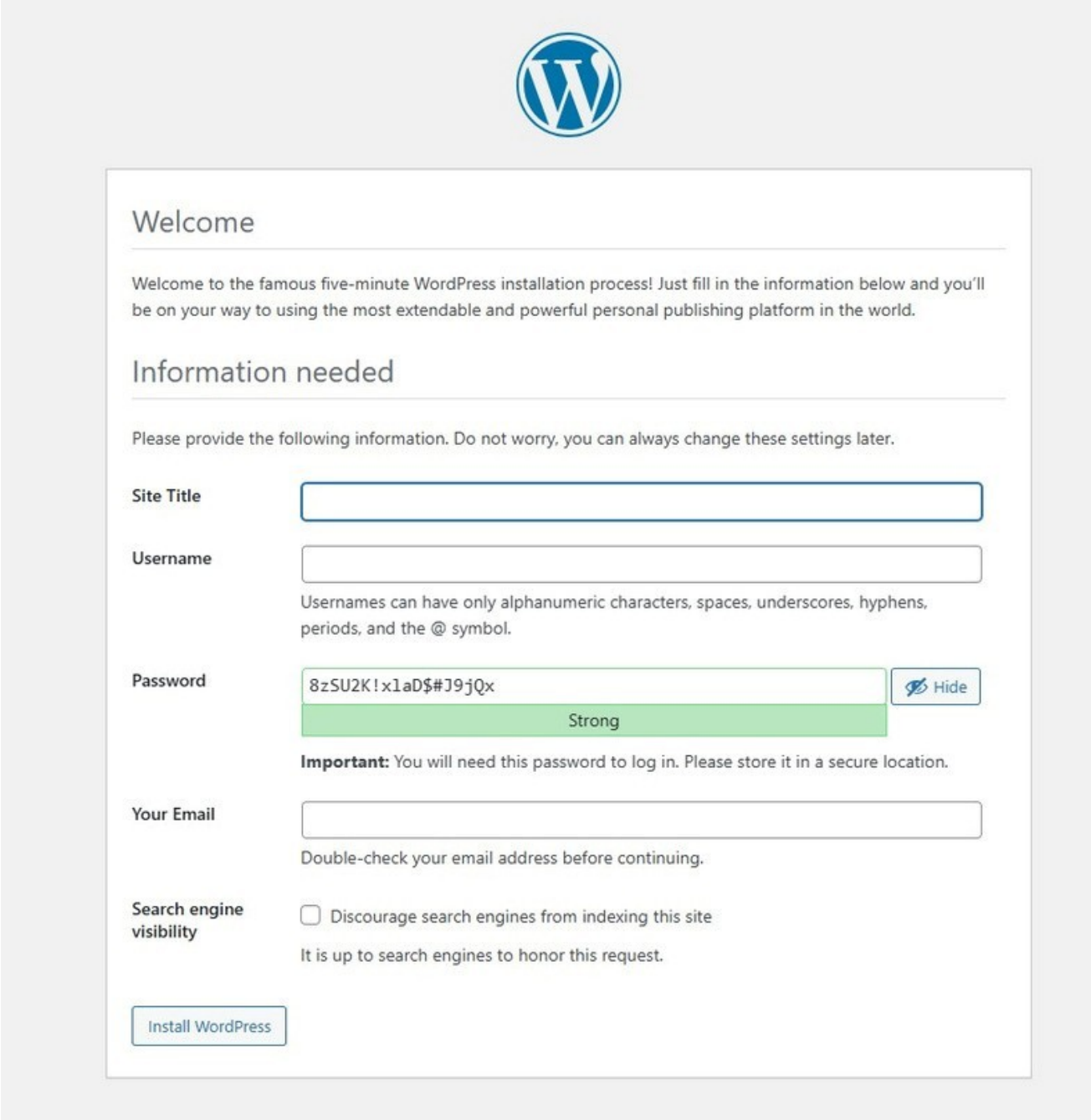
> Observation: Containers are now managed by Swarm, not directly by you.
>
> 
>
>
> —

Task 5: Access WordPress

Open your browser:

```
http://localhost:8080
```

You should see the WordPress setup screen (same as Experiment 6).



Important: Even though Swarm is managing it, the application works the same way!

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Task 6: Scale the Application (Swarm's Superpower)

This is where Swarm shines over plain Compose.

Scale WordPress from 1 to 3 replicas:

```
docker service scale wpstack_wordpress=3
```

Expected output:

```
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
```

```
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service wpstack_wordpress converged
```

Verify scaling:

```
docker service ls
```

Notice the REPLICAS column shows `3/3` for wordpress.

```
ID             NAME                MODE         REPLICAS   IMAGE              PORTS
pr8zglaqgkcy   wpstack_db          replicated   1/1        mysql:5.7
u694ei6iss8a   wpstack_wordpress   replicated   3/3        wordpress:latest   *:8080->80/tcp
```

```
docker service ps wpstack_wordpress
```

You'll see 3 running instances of WordPress.

```
ID             NAME                    IMAGE              NODE             DESIRED STATE   CURRENT STATE               ERROR     PORTS
zn32dqmt579j   wpstack_wordpress.1     wordpress:latest   docker-desktop   Running         Running 5 minutes ago
2sv7qbg4z915   wpstack_wordpress.2     wordpress:latest   docker-desktop   Running         Running about a minute ago
tt8yptk8m06z   wpstack_wordpress.3     wordpress:latest   docker-desktop   Running         Running about a minute ago
```

Check containers:

```
docker ps | grep wordpress
```

You'll see 3 WordPress containers running simultaneously!

```
2b947b7e9f19   wordpress:latest   "docker-entrypoint.s…"   About a minute ago   Up About a minute   80/tcp   wpstack_wordpress.2.2sv7qbg4z9151rkt9w3ur8wdc
d8f74869a315   wordpress:latest   "docker-entrypoint.s…"   About a minute ago   Up About a minute   80/tcp   wpstack_wordpress.3.tt8yptk8m06z2sjkwhgeb9kjh
8e07430009c7   wordpress:latest   "docker-entrypoint.s…"   5 minutes ago        Up 5 minutes        80/tcp   wpstack_wordpress.1.zn32dqmt579j2kt81elvb8ul0
```

What Just Happened?

Before Scaling	After Scaling
1 WordPress container	3 WordPress containers
No load distribution	Swarm balances traffic
Manual scaling needed	One command scaling

Important Question: How do 3 containers share port 8080?

Answer: Swarm creates an internal load balancer.

- All 3 containers receive traffic
- You still access localhost:8080
- Swarm distributes requests automatically

Task 7: Test Self-Healing (Automatic Recovery)

Self-healing = Swarm automatically replaces failed containers.

Step 1: Find a WordPress container

```
docker ps | grep wordpress
```

Copy the CONTAINER ID of one WordPress container.

Step 2: Kill it (simulate a crash)

```
docker kill <container-id>
```

Example: `docker kill`

```
a1b2c3d4e5f6
```

Step 3: Watch Swarm recreate it

```
docker service ps wpstack_wordpress
```

What you'll see:

```
ID             NAME                     IMAGE              NODE     DESIRED STATE   CURRE
xxxxx          wpstack_wordpress.1      wordpress:latest   your-pc   Running         Runn
yyyyy          wpstack_wordpress.2      wordpress:latest   your-pc   Running         Runn
zzzzz          wpstack_wordpress.3      wordpress:latest   your-pc   Running         Runn
aaaaa          \_ wpstack_wordpress.3   wordpress:latest   your-pc   Shutdown        Fail
```

Notice:

- The killed container shows Shutdown/Failed
- A new container is automatically created

  Total replicas still = 3

```
ID             NAME                  IMAGE              NODE             DESIRED STATE   CURRENT STATE                ERROR                       PORTS
zn32dqmt579j   wpstack_wordpress.1   wordpress:latest   docker-desktop   Running         Running 7 minutes ago
4mze32rjtwfm   wpstack_wordpress.2   wordpress:latest   docker-desktop   Running         Running less than a second ago
2sv7qbg4z915    \_ wpstack_wordpress.2   wordpress:latest   docker-desktop   Shutdown        Failed 6 seconds ago         "task: non-zero exit (137)"
tt8yptk8m86z   wpstack_wordpress.3   wordpress:latest   docker-desktop   Running         Running 3 minutes ago
```

Step 4: Verify new container is running

```
docker ps | grep wordpress
```

Still 3 containers running — Swarm fixed it automatically!

> This is self-healing: No manual restart needed.

```
5a6bee0180ec   wordpress:latest   "docker-entrypoint.s…"   3 minutes ago    Up 3 minutes    80/tcp   wpstack_wordpress.2.4mze32rjtwfmel494r8ibvueq
d8f74869a315   wordpress:latest   "docker-entrypoint.s…"   7 minutes ago    Up 7 minutes    80/tcp   wpstack_wordpress.3.tt8yptk8m06z2sjkmhgeb9kjh
8e07430009c7   wordpress:latest   "docker-entrypoint.s…"   10 minutes ago   Up 10 minutes   80/tcp   wpstack_wordpress.1.zn32dqmt579j2kt81elvb8ul0
```

Task 8: Remove the Stack

Clean up everything:

```
docker stack rm wpstack
```

Expected output:

```
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
```

Verify removal: `docker service ls`

(Should show no services)

```
docker ps
```

(WordPress/MySQL containers should be gone)

> Note: Volumes persist unless you delete them manually with `docker volume prune`.

alt text

PART C – ANALYSIS (Compose vs Swarm)

Side-by-Side Comparison

Feature	Docker Compose	Docker Swarm
Scope	Single host only	Multi-node cluster
Scaling	`--scale` flag (basic, no load balancing)	`docker service scale` (built-in)
Load Balancing	No (port conflicts)	Yes (internal LB)
Self-Healing	No (must restart manually)	Yes (automatic)
Rolling Updates	No	Yes (zero downtime)
Service Discovery	Via container names	Via DNS + VIP
Use Case	Development, testing	Simple production clusters
Complexity	Low	Medium

When to Use What?

```
  Development ──────► Compose
Testing       ──────► Compose
  Small Production ─► Swarm
  Large Production ─► Kubernetes
```

PART D – IMPORTANT OBSERVATIONS FOR STUDENTS

Observation 1: Compose File Reuse

Same YAML file works for both Compose and Swarm!

Command	Mode
`docker compose up -d`	Compose (single host, no orchestration)
`docker stack deploy`	Swarm (orchestration enabled)

Key insight: Swarm extends Compose, not replaces it.

Observation 2: Containers vs Services

Concept	Meaning
Container	A single running instance
Service	A definition of how to run containers (image, replicas,

etc.) In Swarm, you manage services, not individual containers.

Observation 3: The Port Mystery Solved

Problem: In Compose, scaling WordPress to 3 would fail because all try to use port 8080.

Solution in Swarm:

- Swarm’s load balancer listens on port 8080 once
- Traffic is distributed to all 3 containers
- internally No port conflicts!

PART E – LEARNING OUTCOME CHECK

Answer These Questions (Write in Your Lab Book)

Q1: Why Compose not enough?

No scaling No self-healing No load balancing

Q2: docker stack deploy vs compose up?

stack deploy → creates services in Swarm compose up → runs containers locally

Q3: How Swarm does self-healing?

Continuously monitors containers Recreates if one fails

Q4: What if docker kill?

Swarm creates a new container automatically

Q5: Same file usable?

Yes, Swarm extends Compose

PART F – OPTIONAL: Multi-Node Swarm (Advanced)

If you have access to multiple VMs or computers on the same network:

On manager node, get join token:

```
docker swarm join-token worker
```

On worker node, join the cluster: `docker`

```
swarm join --token <token> <manager-ip>:2377
```

Verify from manager:

```
docker node ls
```

Now your stack will distribute containers across multiple machines automatically!

Summar y

You started with	You can now do
Single container (`docker run`)	Multi-container (Compose)
Manual scaling	One-command scaling (`scale`)
Manual recovery	Automatic self-healing
Single host	Multi-host cluster ready

Final Takeaway

> Compose defines the application. Swarm runs it reliably.

Quick Reference Card